

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1
Subject: 353152 Electronic Circuits I
Date: 9 December 2015

Year: 2015
Section: 5 - 6
Time: 13.00 - 15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Text books and Dictionaries are **NOT** allowed.
 4. A calculator is allowed.
 5. Electronic communication devices are **NOT** allowed in the examination room.
 6. The examination has 9 pages (including this page), 2 parts (35 questions) and a total score of 110 points.
 7. Write all your answer on this examination sheet.
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Part 1 (60 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. For the dc analysis the network can be isolated from the indicated ac levels by replacing the capacitor with _____.
 (A) an open circuit equivalent (B) a short circuit equivalent
 (C) a source voltage (D) a source current
2. In a fixed-bias circuit, the slope of the dc load line is controlled by _____.
 (A) R_B (B) R_C (C) V_{CC} (D) V_{BE}
3. The _____ the stability factor, the _____ sensitive the network is to variations in that parameter.
 (A) higher, more (B) higher, less
 (C) lower, more (D) lower, less
4. In any amplifier employing a transistor, the collector current I_C is sensitive to _____.
 (A) β (B) V_{BE} (C) I_{CO} (D) All of the above
5. In a transistor-switching network, the operating point switches from _____ to _____ regions along the load line.
 (A) cutoff, active (B) cutoff, saturation
 (C) active, saturation (D) None of the above
6. _____ is less dependent on the transistor beta.
 (A) Fixed bias (B) Emitter bias
 (C) Voltage divider (D) Voltage feedback
7. Which type of amplifier uses pulse (digital) signals in its operation?
 (A) Class A (B) Class B or AB (C) Class C (D) Class D
8. Which of the following voltages must have a negative level (value) in any npn bias circuit?
 (A) V_{BE} (B) V_{CE} (C) V_{BC} (D) V_{CQ}
9. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?
 (A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) Undefined
10. The saturation region is defined by V_{CE} _____ V_{CEsat} .
 (A) $>$ (B) $<$ (C) $\leq$ (D) $\geq$
11. _____ is the least stabilized circuit.
 (A) Fixed bias (B) Emitter-stabilized bias
 (C) Voltage divider (D) Voltage feedback
12. For the typical transistor amplifier in the active region, V_{CE} is usually about _____ % to _____ % of V_{CC} .
 (A) 10, 60 (B) 25, 75 (C) 40, 90 (D) 50, 50

13. Changes in temperature will affect the level of _____.
 (A) current gain β
 (B) leakage current I_{CO}
 (C) both current gain β and leakage current I_{CO}
 (D) current gain β , leakage current I_{CO} and voltage V_{BE}
14. For a transistor operating in the saturation region, the collector current I_C is at its _____ and the collector-emitter voltage V_{CE} is to the _____.
 (A) minimum, left of the V_{CEsat} line
 (B) minimum, right of the V_{CEsat} line
 (C) maximum, left of the V_{CEsat} line
 (D) maximum, right of the V_{CEsat} line
15. Which class of power amplifiers conducts through 360° of input?
 (A) class A (B) class B (C) class AB (D) class C
16. Which class of power amplifiers has Q-point at the cut-off point?
 (A) class A (B) class B (C) class AB (D) class C
17. Which class of power amplifiers that conducts less than 180° of input?
 (A) class A (B) class B (C) class AB (D) class C
18. How can the class A amplifier efficiency be improved from 25%?
 (A) Increase input current. (B) Reduce input current.
 (C) Add transformer at input side. (D) Add transformer at output side.
19. What is the benefit of using Darlington pair and feedback pair transistors for push-pull operation.
 (A) Reduce output swing (B) Increase output power
 (C) Increase input current (D) Reduce distortion
20. Which type of following power amplifiers has the highest efficiency?
 (A) class A (B) class B (C) class AB (D) class C
21. What is the cause of crossover distortion in push-pull circuit?
 (A) Two transistors do not turn on/off simultaneously.
 (B) Heat at transistor.
 (C) Two transistors are not the same type (pnp and npn).
 (D) Input current is the sine wave.
22. What is the value of I_{dc} of class B amplifier if peak output current $I_L(p)$ is 10A.
 (A) 0.318 A. (B) 0.636 A. (C) 0.707 A. (D) 1.414 A.
23. What is the value of $P_o(ac)$ of class B amplifier if $V_L(p-p) = 10$ V and $R_L = 2 \Omega$.
 (A) 50 W (B) 25 W (C) 6.25 W. (D) 5.25 W.

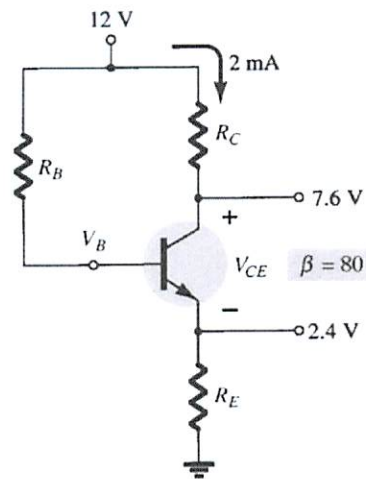


Figure 1.

24. From Figure 1, determine R_E

- (A) $1\text{ k}\Omega$ (B) $1.2\text{ k}\Omega$ (C) $4.8\text{ k}\Omega$ (D) $6\text{ k}\Omega$

25. From Figure 1, determine V_{RB}

- (A) 8.9 V (B) 9.6 V (C) 11.3 V (D) 12 V

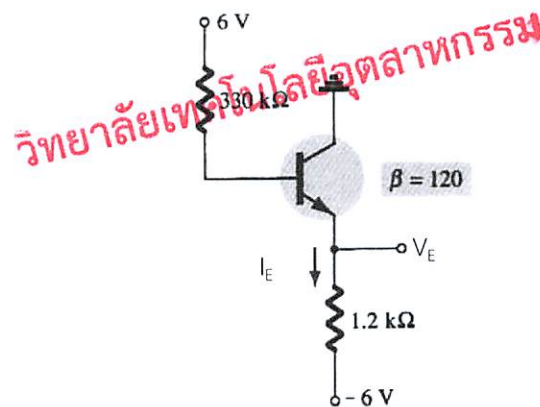


Figure 2.

26. From Figure 2, find the current I_E .

- (A) $2.87\text{ }\mu\text{A}$ (B) 2.87 mA (C) $4.12\text{ }\mu\text{A}$ (D) 4.12 mA

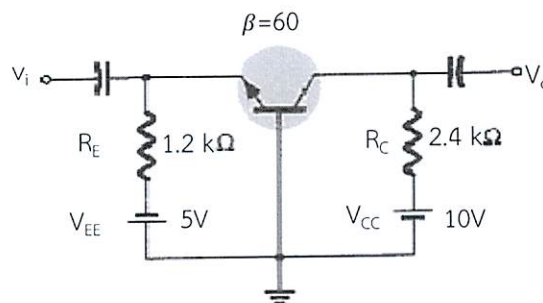


Figure 3.

27. Determine the values of V_{CB} and I_B for the circuit in Figure 3.

- (A) 1.4 V , $59.7\text{ }\mu\text{A}$ (B) -1.4 V , $59.7\text{ }\mu\text{A}$
(C) -9.3 V , $3.58\text{ }\mu\text{A}$ (D) 9.3 V , $3.58\text{ }\mu\text{A}$

28. For a properly biased *pnp* transistor, let $I_C = 10 \text{ mA}$ and $I_E = 10.2 \text{ mA}$. What is the level of I_B ?

- (A) 0.2 A (B) 200 mA (C) 200 μA (D) 20.2 mA

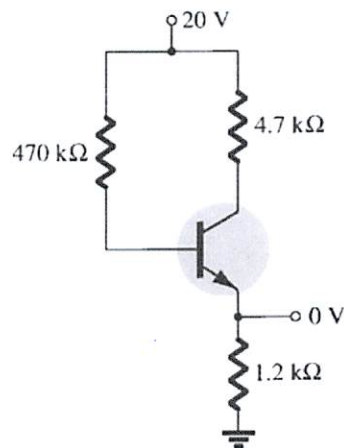


Figure 4.

29. From Figure 4, the measurement of $V_E = 0 \text{ V}$ dose not function correctly. What could NOT be the problem?

- (A) Open-circuit in the base circuit
(B) Bad connection of emitter terminal
(C) Damaged transistor
(D) Short base-emitter junction

30. Which of the following is NOT the application of a transistor?

- (A) Amplification of signal (B) Switching and control
(C) Computer logic circuitry (D) Air conditioner

Part 2 (50 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

1. From Figure 5, calculate

- (a) $P_o(\text{ac})$
(b) $P_i(\text{dc})$
(c) $\% \eta$

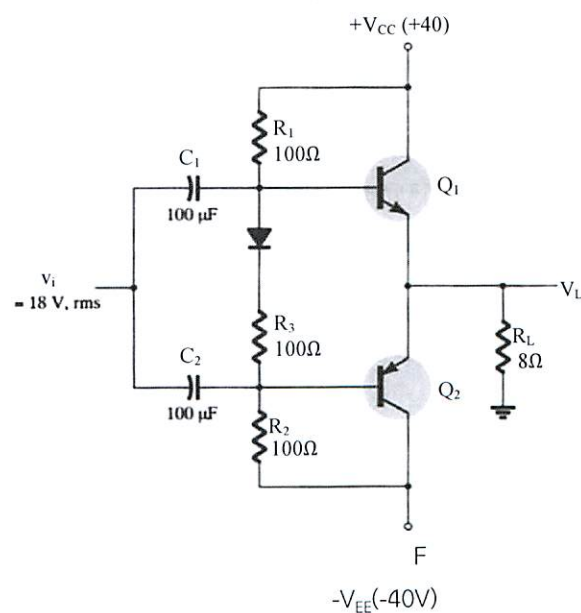


Figure5.

2. From Figure 6, determine the level of V_{CE} , I_C and I_B .

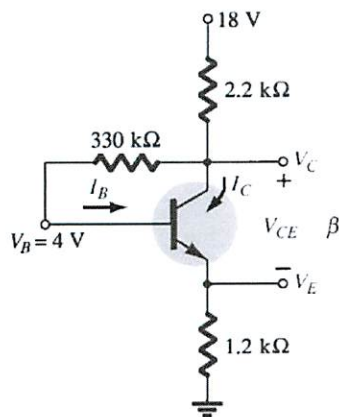


Figure 6.

3. From Figure 7, determine I_B , I_C , and V_{CE} with $\beta = 100$

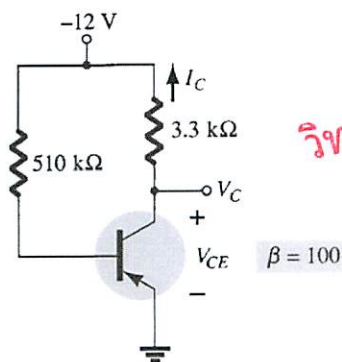


Figure 7.

4. Use the table below to determine the change I_C in from 100°C to 25°C for the transistor defined in this table for emitter-bias with $R_B = 240 \text{ k}\Omega$, $R_E = 2 \text{ k}\Omega$ and $\beta = 100$ due to the $S(I_{CO})$ and $S(V_{BE})$ stability factors.

$$S(I_{CO}) = (\beta + 1) \frac{1 + R_B/R_E}{(\beta + 1) + R_B/R_E} \text{ and } S(V_{BE}) = \frac{-\beta}{R_B + (\beta + 1)R_E}$$

$$\Delta I_C = S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE}$$

T (°C)	I_{CO} (nA)	β	V_{BE} (V)
-65	0.2×10^{-3}	20	0.85
25	0.1	50	0.65
100	20	80	0.48
175	3.3×10^3	120	0.3

5. Determine I_C , V_{CE} and I_B from Figure 8 by using the exact (Thevenin) approach.

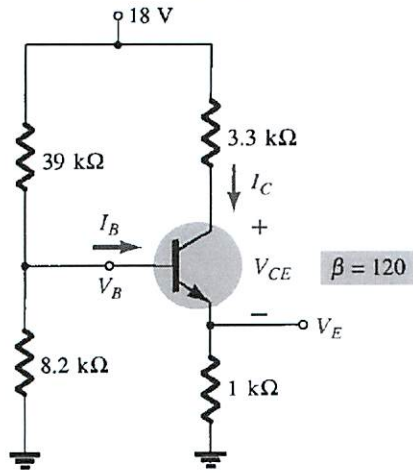
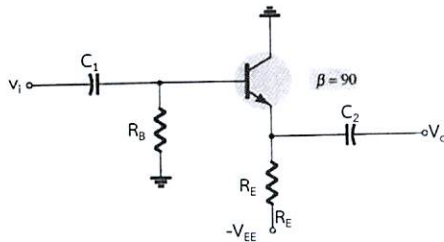


Figure 8.

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Handout

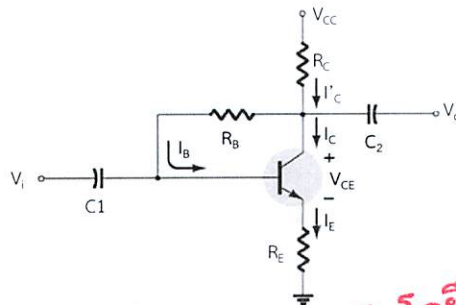
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

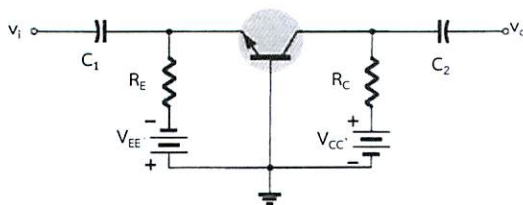
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_o(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\%h = 50 \left(\frac{V_{CE(max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(min)}} \right)^2$$

$$\frac{R_L}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

Class B amplifier

$$P_o(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC} \left(\frac{2}{\pi} I_L(p) \right)$$

Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

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